Q1. Rankine Cycle Theory: Explain the Rankine cycle with a detailed schematic and T-s diagram. Find efficiency equation

Q2. A reverse Rankine cycle is used as a refrigeration cycle. The refrigerant enters the compressor at a pressure of 0.3 MPa and a temperature of 5°C and is compressed to a pressure of 1.2 MPa. The enthalpy at the compressor inlet is 240 kJ/kg, and at the outlet, it is 280 kJ/kg. The refrigerant leaves the condenser at 1.2 MPa with an enthalpy of 110 kJ/kg and expands in the expansion valve to 0.3 MPa with an enthalpy of 105 kJ/kg. Calculate:

I. **Work input to the compressor (Wcomp)**

II. **Heat rejected by the condenser (Qout)**

III. **Heat absorbed by the evaporator (Qin)**

V. **Coefficient of performance (COP) of the cycle**